Mathematical paragraph continuity

The observation history $h_t = (o_{t+1-H}, a_{t+1-H}, ..., o_t)$ maps to an action $a_t \sim \pi_{\theta(\cdot \mid l, h_t)}$, where $H > 1$ is the temporal horizon and the instruction is denoted by l. Each observation $o_t = (v_t, p_t)$ represents visual and proprioceptive information. These definitions belong to one continuous paragraph.

The static task is described by $(e_i, x_i, q_i, G, \Phi, \Psi)$, where x_i is the initial position and q_i is the initial orientation. A state predicate $\varphi : S \times A \to \{0, 1\}$ and a trajectory predicate $\psi : H \to \{0, 1\}$ remain inline parts of this complete explanation.

Formally, we aim to solve

$$\pi^\star = \arg \max_{\pi_\theta \in \Pi_C} J(\pi_\theta) \tag{1}$$

The numbered expression is independently centered and its star belongs to it.

Two-column evidence

The first column describes $x_i^2 + y_i^2$ as a useful quantity and contains a continuous explanation with ordinary words around the mathematics. The symbols must not become isolated untranslated lines.

$$E = mc^2 \tag{2}$$

The expression above remains a display equation.

The next paragraph contains $\alpha + \beta = \gamma$ in the middle of its explanation. This text should retain its mathematical spans without merging distinct paragraphs.